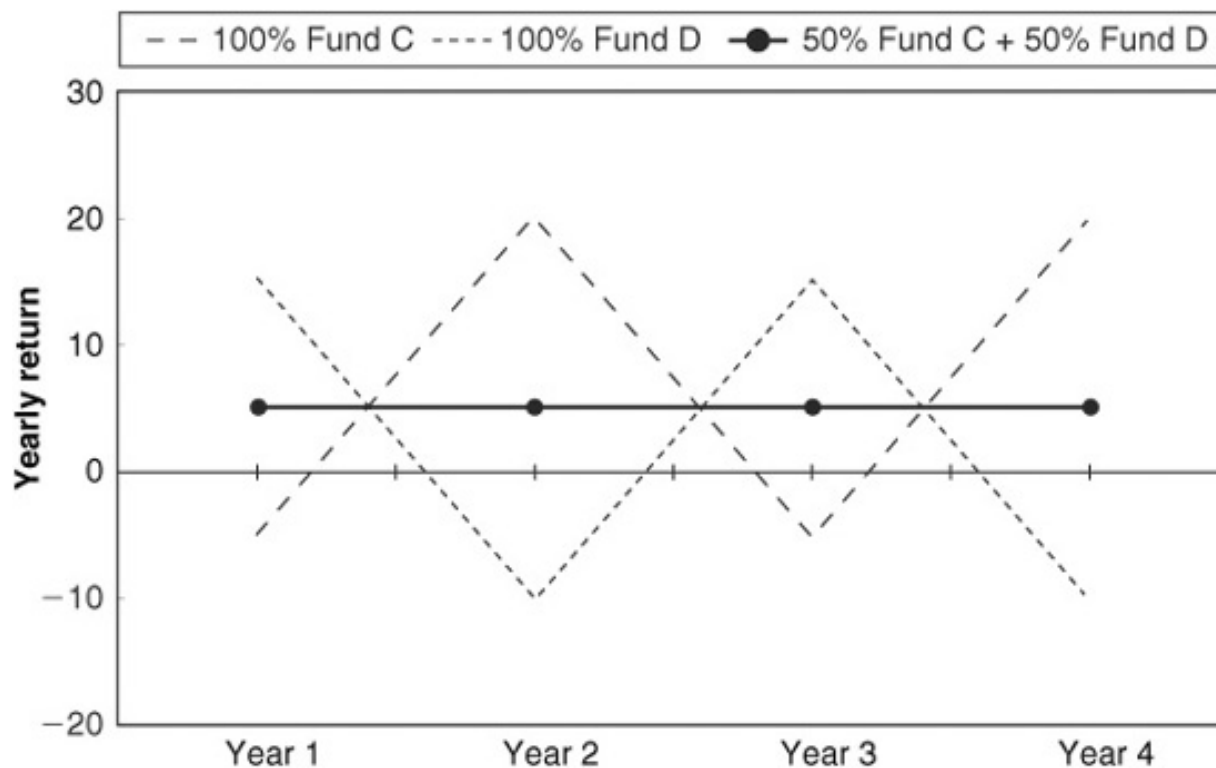




Correlation is measured using a range between +1 and -1. Two investments that have a correlation of +0.3 or greater are considered positively correlated. When two investments have a correlation of -0.3 or less, this is considered negative correlation. A correlation coefficient between +0.3 and -0.3 is considered non-correlated.

When two investments are noncorrelated, either the movement of one does not track the movement of the other or the tracking is inconsistent and shifts between positive and negative. [Figure 3-4](#) represents two investments that are noncorrelated; sometimes they move together, and sometimes they do not. There is a diversification benefit from investing in noncorrelated assets. Typically, the best asset class pairs that you will be able to find have noncorrelation, or they may have positive correlation at times and negative correlation at other times.

[Table 3-2](#) is a summary of the diversification benefits from correlation. The table assumes that all three portfolios have a simple average return of 5 percent per year, although they have different compounded returns because of the volatility of each portfolio. Portfolio 1 held two investments with negative correlation, and it produced the lowest risk and the highest return. In contrast, Portfolio 3 held two investments with positive correlation. That portfolio had more risk and achieved the lowest return.